

GunnerAgent: A Two-Layer Negotiation Agent for the ANAC 2026 Human-Agent Negotiation League

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Background & Objectives

ANAC 2026's HAN League: agent bargains with a human under Alternating Offers, with free-text messages.

~10 offers per side in 5 minutes; the partner's utility function is hidden.

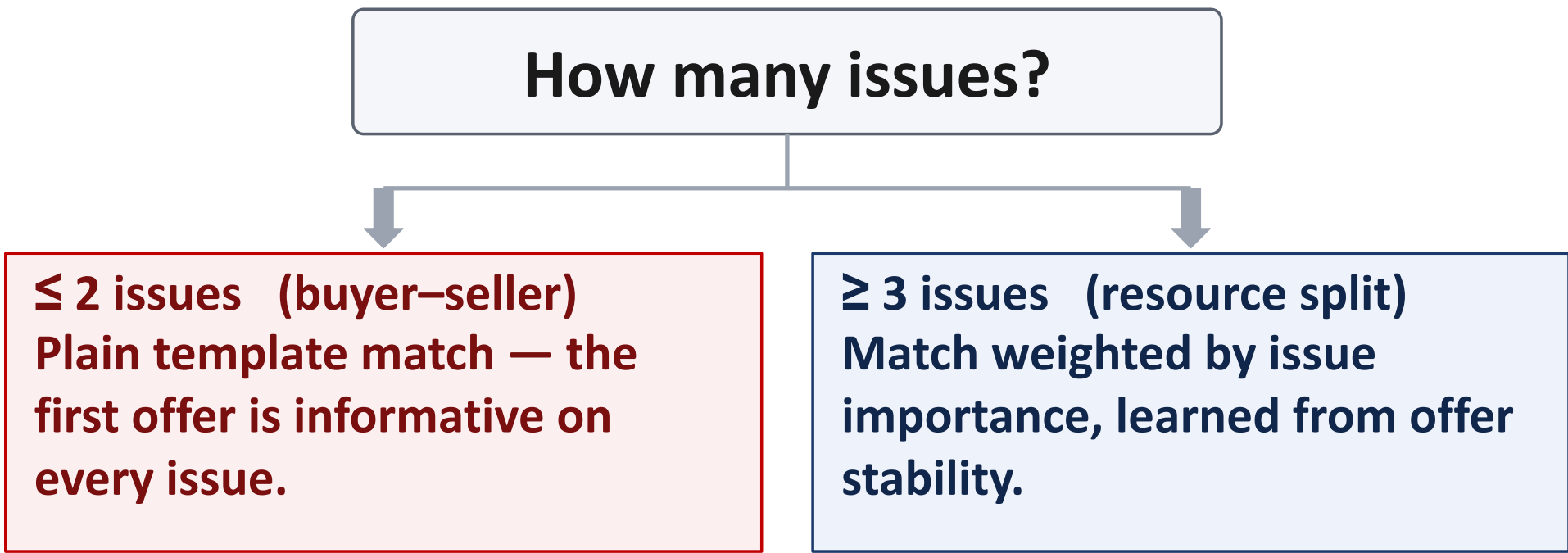
Scored on utility won and on the partner's perception of the agent.

Objective: extract value in ~10 rounds, sound human, stay reproducible and LLM-independent.

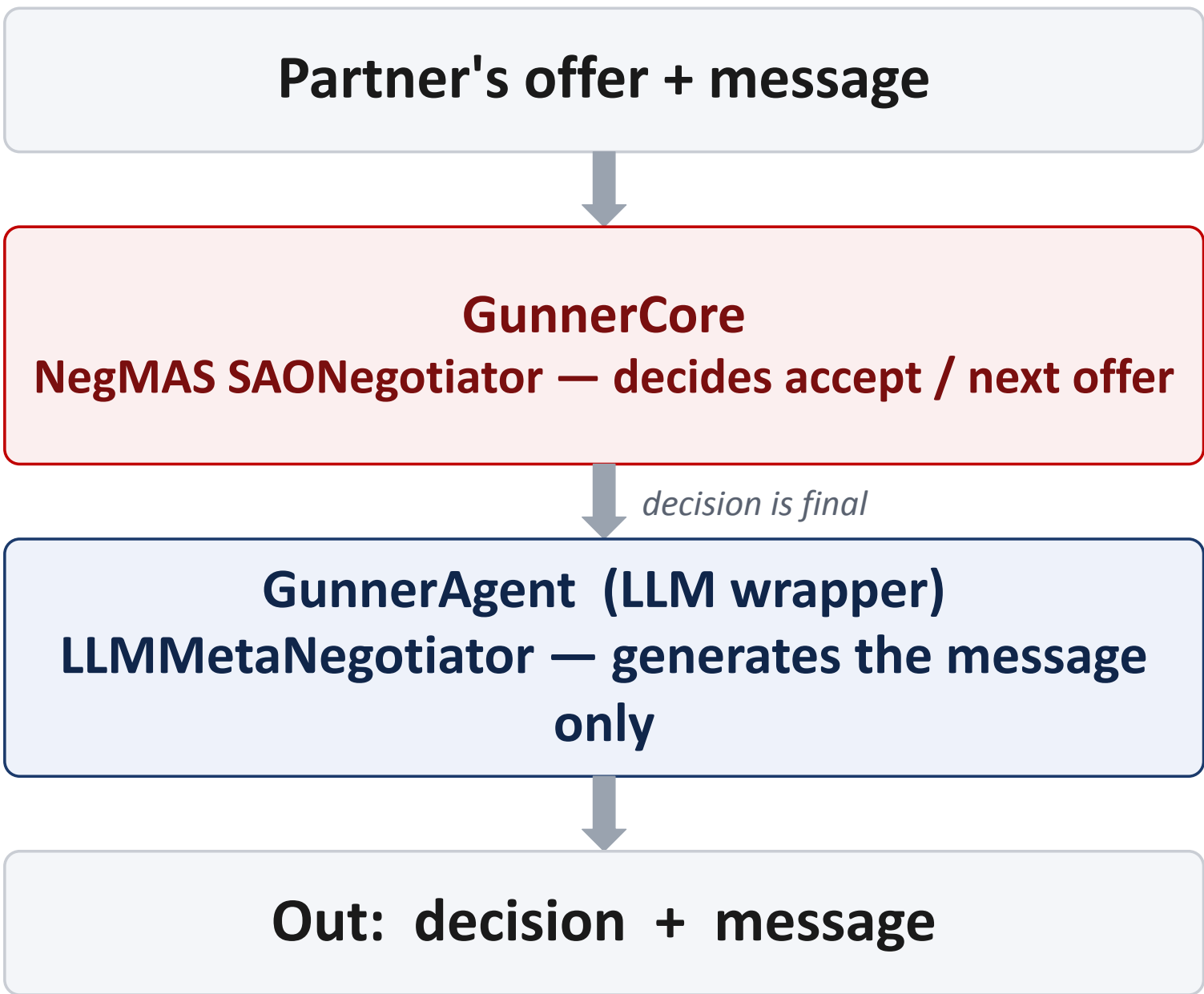
Adaptive Opponent Model

Frequency models fail: HardHeaded, CUHK, Smith need hundreds of rounds.

What works: treat the partner's first offer as a preference template — undistorted by concession.



Two-Layer Architecture



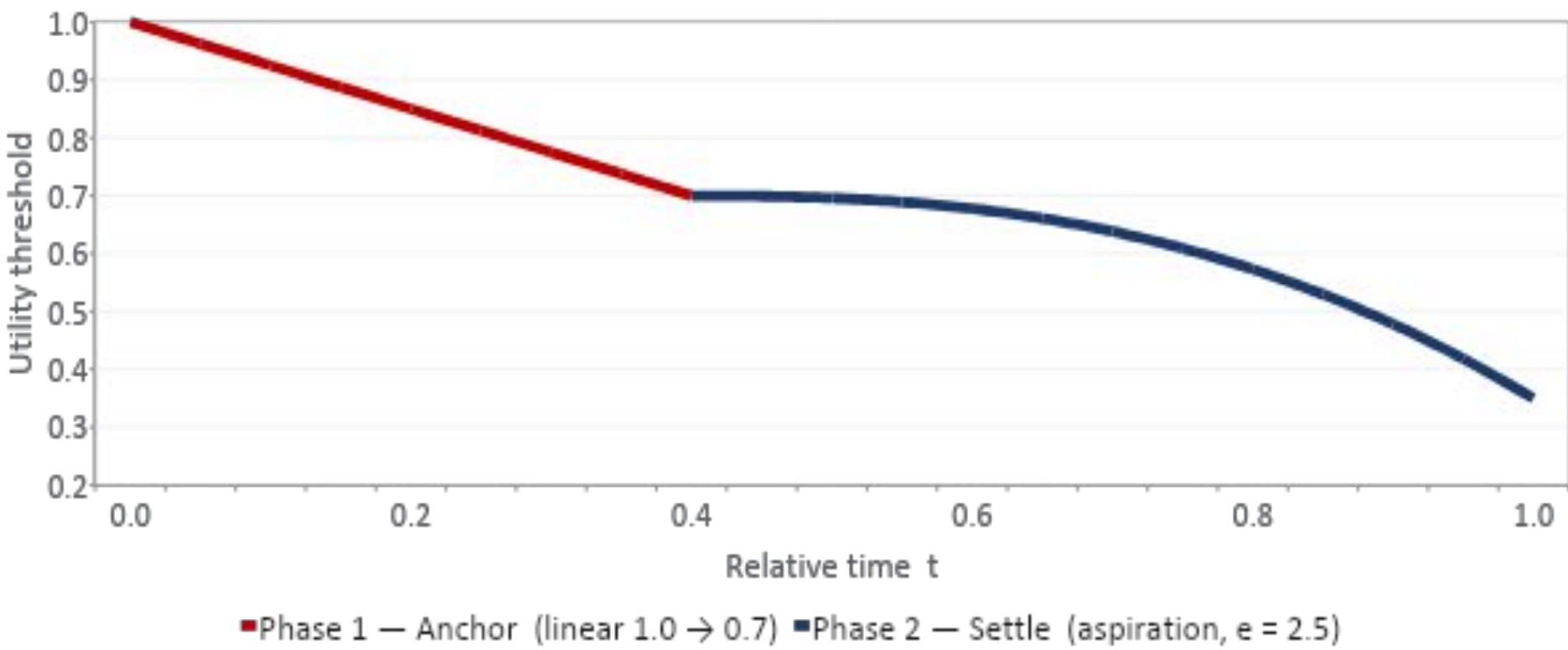
LLM Message Layer - Modular Prompt

- Presentation Only:** The LLM only generates the text message; the numerical core strictly decides offers and acceptances.
- Modular Prompt Assembly:** Built from independent functions for precise prompt control without instruction interference.
- Move & Partner Classification:** Maps every turn to specific move labels to match message tone directly with negotiation behavior.
- Hidden information:** the model sees only a verbal description — never utilities, reservation value, or priorities.
- Fallback:** on timeout or malformed JSON, a fixed line ships with the move — the agent is never silent.
- Single-Point Clarity:** Eliminates conflicting instructions while using a tuned temperature (0.7) to prevent paraphrasing errors and truncation.

Two-Phase Concession Curve

$$aspiration(t) = (M - rv)(1 - t^e) + rv$$

No single exponent survives ~10 turns: Boulware is robotic, Conceder gives away value at once, linear lets the partner wait for the last turn.



Phase 1 - Anchor (t < 0.4)
Linear 1.0 → fixed 0.7 floor. An rv-scaled floor froze the agent near 1.0.

Phase 2 - Settle (t ≥ 0.4)
Aspiration curve, e = 2.5, 0.7 → max(1.5·rv, 0.35). Holds, then concedes sharply.

Parameter	Value	Why
t_split	0.4	Anchors clearly, still leaves rounds to bargain.
Phase-1 floor	0.7	Highest value a human still reads as a real concession.
e (Phase 2)	2.5	Neither robotic repetition nor premature concession.
Base floor	max(1.5·rv, 0.35)	Secures a profit over rv; keeps an agreement zone.
max_repeats	5	Caps identical proposals so the agent stays dynamic.

Note: the report gives the Phase-2 floor as 0.3 in prose but max(1.5·rv, 0.35) in its table; the table value is used here.

Conclusions & Key Takeaways

- Splitting the numerical core from the LLM:** full control and prevents LLM strategy glitches.
- Two phases beat one curve** in a ~10-round human budget
- Modular, situation-controlled prompt** keeps the LLM generator on point.
- Opponent RV is hard to predict** — model an RV range instead of one value.